

NeuroFleets Internship Program

Engineering internships in autonomous robot fleets, edge AI, and multi-agent learning



Who We're Looking For

What matters most to us is that you're genuinely interested and willing to learn. You don't need to tick every box under a track to apply.

If you have solid fundamentals in engineering or computer science, a real interest in robotics and AI, and you're happy to pick up new tools as you go, we'd like to hear from you.

TRACK 1

Reinforcement Learning Intern (Multi-Agent Fleet Learning)

Role. You'll run and analyse multi-agent RL training experiments for our robot fleet pipelines, improve the simulation environments, build tools to evaluate results, and help track down problems in training runs.

What we're looking for

- Comfortable with Python (PyTorch and NumPy a plus)
- Able to read learning curves and work out what's happening
- The patience to sit with long training runs and debug them
- Some grounding in reinforcement learning, or keen to pick it up
- A plus: multi-agent RL (IPPO, MAPPO, QMIX, CTDE), meta-RL, reward shaping and curriculum learning, or experiment tracking with TensorBoard / Weights & Biases

TRACK 2

Robotics Integration & Commercial Deployment Intern (ROS, Build / Deploy / Test)

Role. You'll help get our software and models running on real customer hardware, and dig into the edge cases and field failures that come up along the way.

What we're looking for

- Comfortable working in Linux and on the command line
- Good Python scripting skills
- Willing to get hands-on with physical hardware and work through tricky problems
- Some ROS or ROS 2, or keen to learn it
- A plus: Docker, more than one ROS distribution, CI/CD pipelines, or deployment on Google Cloud Platform (GCP)

TRACK 3

AI Platform & Curriculum Intern (NeuroFleets Academy)

Role. You'll build AI agents that check students' hands-on robotics skills as they work through curated video learning paths, so course certification can happen automatically.

What we're looking for

- Comfortable with Python, ideally some backend work (FastAPI or similar)
- Interested in LLMs and prompt engineering
- Happy to build automated grading pipelines and lay out clear learning paths
- A plus: React / TypeScript, RAG or agentic workflows, some ROS 2 / MARL / robotics background, or Supabase and authentication

TRACK 4

FleetOS Core Systems Intern

Role. You'll work on the core backend that handles multi-robot dispatching, task allocation, and path execution across our mobile fleets, keeping it fast and reliable.

What we're looking for

- Strong Python, and some C++ (modern C++14/17 ideal)
- A good handle on algorithms and data structures (graph search, queues, optimisation)
- Careful attention to detail with real-time data and concurrency
- Familiarity with REST APIs or FastAPI, or keen to learn
- A plus: search-based path planning, dynamic routing for AMRs, MQTT / WebSockets, or Docker

TRACK 5

Drone Autonomy Intern (ArduPilot / Gazebo / Embedded)

Role. You'll build, simulate, and deploy autonomous drone control pipelines on embedded hardware (Raspberry Pi) for farmland monitoring and precision agriculture.

What we're looking for

- Good Python or C++ skills
- Keen on hardware, field deployment, and flight testing
- Some experience with a Raspberry Pi, or comfortable getting into it
- Some experience with the ArduPilot / Pixhawk stack and Gazebo simulation, or keen to learn
- A plus: ROS 2 / MAVROS bridges, or building custom Gazebo worlds (e.g. agricultural terrain)

TRACK 6

Cloud Build, Model Security & Edge Deployment Intern

Role. You'll build cloud-native GCP pipelines that compile, secure (model encryption and TPM-based hardware security), and deploy multi-architecture software (arm64/aarch64 and x86_64) across ROS distributions, straight to remote customer robots.

What we're looking for

- Comfortable in Linux: command line, shell scripting (bash), and remote admin (SSH, systemd)
- Some Python backend work (FastAPI) to orchestrate deployments
- A basic understanding of cloud services, ideally GCP (Cloud Build, Artifact Registry, GCS, Cloud Run / Compute Engine)
- Some ROS / ROS 2 and Ubuntu familiarity, or keen to learn
- A plus: hardware security (TPM, model encryption, secure boot), managing remote edge devices, or React / TypeScript dashboards

Interested in more than one track? We'd encourage you to apply for several.

Position: Unpaid. Part-time or full-time. Remote, or in person in Jaffna.